

ASIF AHAMED S

Embedded Systems Developer

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Summary

Embedded Systems Developer who learns by building, debugging, and documenting. Comfortable prototyping with ESP32 and STM32 on perfboards, with experience in custom PCB design and wireless protocols such as ESP-NOW. Currently developing bare-metal STM32 firmware by reading reference manuals to write custom, register-level device drivers for UART and I2C, while expanding into FreeRTOS and building static-analysis tooling for C-code Verification & Validation (V&V). Built a college-funded 10-node wireless buzzer system (20 ms latency) and won 2nd Place, Hardware at Makeathon 7.0. Seeking a full-time or long-term internship role to apply strong C programming, device-driver development, and hardware-software integration skills in production-grade, reliability-focused firmware.

Education

Bachelor of Engineering — Electronics and Communication Engineering (ECE) **CGPA: 8.3/10**
Rajalakshmi Engineering College, Chennai Expected Graduation: 2027

Higher Secondary (XII) — PCMB **Percentage: 93.8%**
Calavala Cunnun Chetty's Hindu Matriculation Higher Secondary School, Tiruvallur 2023

Technical Skills

- **Languages:** C (pointers, structs, memory management — Stack/Heap/Static, bitwise operations, `volatile`), Python (Competitive Programming), MATLAB
- **Embedded Platforms:** ESP32 (Arduino Framework), STM32 (Bare-metal register-level & HAL), Arduino Uno
- **RTOS & OS:** FreeRTOS (tasks, queues, semaphores, mutexes), Linux CLI, Linux System Programming (processes, threads, IPC)
- **Protocols:** I2C, SPI, UART, ESP-NOW, WiFi, BLE, LoRa
- **Hardware & Testing:** Multimeter, Oscilloscope, Logic Analyzer, SWD Debugger, Perf-Board Prototyping, Signal-Integrity Verification
- **Tools & Workflow:** PlatformIO, VS Code, Arduino IDE, STM32CubeIDE, Git (version control & branching), Make/CMake, GitHub Actions (CI/CD)
- **Core Concepts:** Interrupt handling, signal debouncing, register-level programming, device drivers, FSM design, Watchdog Timers, low-power design, hardware-software co-design, Verification & Validation (V&V), test automation, fault diagnosis

Projects

Wireless Quiz Buzzer System — College-Funded Project *ESP32 · ESP-NOW · WebSockets*
GitHub: github.com/asifahamed-ece/wireless-quiz-buzzer

- Architected a 10-node ESP-NOW quiz system with interrupt-driven button handling, achieving a 20 ms response time.
- Built a real-time web dashboard (EspAsyncWebServer + WebSockets) with LittleFS for persistent score/battery updates; deployed for college quiz events and recognized on-stage before 800+ attendees.
- Authored complete hardware-setup and system documentation on GitHub for reproducibility.

Bare-Metal STM32 I2C Driver (SSD1306 OLED) *STM32F411 · C · Register-Level · No-HAL*
GitHub: github.com/asifahamed-ece/BareMetal_I2C_Driver

- Authored a from-scratch, register-level I2C driver for the STM32F411 Black Pill (no HAL) to drive an SSD1306 OLED.
- Configured RCC clock trees, GPIO alternate functions, and I2C peripheral registers directly from the ARM Cortex-M4 reference manual.
- Debugged signal integrity and bus timing using an oscilloscope to ensure robust hardware-software integration.

CiOpt — C Code V&V & Optimization Engine *C · Tree-sitter · Static Analysis · GitHub Actions*
GitHub: github.com/asifahamed-ece/ciopt

- Developed a compiler-inspired static-analysis tool that estimates Big-O complexity and detects bottlenecks, anti-patterns, and dead code in C — without compiling or running the code.
- Engineered detection for unsafe memory operations (`malloc/free` in hot paths, unsafe string functions) and missing null checks, targeting common embedded-C defects.
- Integrated as a CI/CD quality gate (GitHub Actions) and VS Code extension, automating V&V and generating terminal/HTML/JSON defect reports.

- Led hardware/firmware for a bidirectional BLE-LoRa mesh SOS node; delivered in 24 h under Rs.1,800/node using ESP32, an LM2596 power circuit, and dual 18650 cells.
- Bridged a Flutter app (BLE) to the LoRa mesh (SF7/SF12) and optimized duty cycling for low-power operation; ranked #2 in Hardware among 55 teams.

Industrial Training & Exposure

Research Trainee — CSIR-CSIO, Chennai

June–July 2025 (2 Weeks)

- Analyzed on-grid solar-inverter testing workflows, including efficiency calculation, MPPT analysis, and waveform-quality assessment using oscilloscopes and power instrumentation.
- Observed ML pipelines (CNN/ANN training on TensorFlow/Keras) and structured test-data documentation practices.

Short-Term Industrial Training — Voltech Manufacturing Company Ltd., Chennai

January 2025 (1 Week)

- Gained hands-on exposure to industrial QA workflows during assembly and testing of Boost-cum-Float Battery Chargers.
- Conducted continuity/voltage checks and learned ESD safety protocols and documentation standards for power-electronics production.

Achievements

- **2nd Place, Hardware Category** — Makeathon 7.0, SVCE

April 2026
- **College-Level Recognition** — Wireless Quiz Buzzer System presented on-stage to 800+ attendees

January 2026
- **Qualified Final Round**, Embedded Programming Challenge (Shaastra, IIT Madras)

January 2025
- **Participant** — Smart India Hackathon (SIH 2024 & SIH 2025)
- **Event Coordinator** — REVTRONICS 2025, ECE Department Technical Symposium

Certifications & Coursework

- Embedded IoT & Industrial Automation (Level 1 & 2) — Taras Systems

April 2025
- Deep Learning Onramp — MathWorks (Self-Paced)

October 2025
- Core MATLAB Skills — MathWorks (Learning Path)

July 2025
- Electronic Systems Design: Hands-On Circuits and PCB Design — NPTEL (Score: 81/100)

October 2024
- Sensors and Actuators — NPTEL (Score: 73/100)

April 2025

Interests

Embedded Systems · IoT & Wireless Networks · Low-Power Design · Computer Networking · Linux CLI Workflows · Home Labbing · Static Analysis & Code Quality